

Statistics and Probability Theory

September 21, 2026

Objectives

Data presentation is the method by which people summarize, organize and communicate information using a variety of tools, such as diagrams, distribution charts, histograms and graphs.

Objectives

- It condense the raw data into a form suitable for statistical analysis.
- It remove the complexities and highlights the feature of data.
- It facilitates comparison and drawing conclusion from data.
- It helps in statistical analysis by separating elements of data set into homogeneous groups and hence bring out the points of similarity and dissimilarity.

Tabular methods

- 1 Frequency Distribution.

Tabular methods

- 1 Frequency Distribution.
- 2 Cumulative Frequency Distribution.

Tabular methods

- ① Frequency Distribution.
- ② Cumulative Frequency Distribution.
- ③ Percentage Frequency Distribution.

Presentation of Quantative Data

Tabular methods

- 1 Frequency Distribution.
- 2 Cumulative Frequency Distribution.
- 3 Percentage Frequency Distribution.
- 4 Cross Tabulation (Contingency Table).

Graphical methods

- 1 Histograms.

Presentation of Quantative Data

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Graphical methods

- ① Histograms.
- ② Frequency Polygon

Presentation of Quantative Data

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Graphical methods

- ① Histograms.
- ② Frequency Polygon
- ③ Frequency Curve

Data Set 1

The following data represent the final result of 19 students of MTH262.

S, No	Gen	Total	LG	GP	S, No	Gender	Total	LG	GP
1	M	71	B	3	11	M	57	D+	1.3
2	M	57	D+	1.3	12	M	64	C+	2.33
3	M	65	C+	2.33	13	M	54	D+	1.3
4	M	75	B+	3.33	14	M	81	A-	3.66
5	F	66	C+	2.33	15	M	75	B+	3.33
6	F	50	D	1	16	F	69	B-	2.66
7	F	68	B-	2.66	17	M	68	B-	2.66
8	M	51	D	1	18	M	68	B-	2.66
9	M	76	B+	3.33	19	M	57	D+	1.3
10	F	87	A	4					

Example-1

Present the data of **Total Marks** given in **Data Set 1** by Frequency Distribution, Histogram, Frequency Polygon, and Frequency Curve.

Frequency Distribution

A frequency distribution is a table that distributes a data set into a suitable number of categories (classes). Data presented in the form of a frequency table are called grouped data

Steps for Frequency Distribution

- 1 Decide on the number of non overlapping groups or classes (K). H.A Sturges proposed a rule,
$$K=1+3.22 \log N$$
$$K=1+3.22\text{LOG}(19)=5.11 \approx 5$$
- 2 Determine the class width or interval. Class Interval(W)=Maximum-Minimum/K = $\frac{87-50}{5} = 7.11 \approx 7$
- 3 Determine where to locate the class limit.
we start classes from 50.
- 4 Distribute the data into appropriate classes.
- 5 Make frequency column

Frequency Distribution: Example-1

Total Marks: 71, 57, 65, 75, 66, 50, 68, 51, 76, 87, 57, 64, 54, 81, 75, 69, 68, 68, 57

Table: Frequency Distribution

Classes	f	%f
50-56	3	15.79
57-63	3	15.79
64-70	7	36.84
71-76	4	21.05
77-83	1	5.26
84-90	1	5.26

Histogram

A histogram is a specialized column graph used in statistics to visualize the underlying frequency distribution of a continuous numerical variable.

- 1 A histogram provides an instant snapshot of how data points spread across a range. It quickly reveals whether the dataset is symmetrical (normal distribution), skewed to the left/right, or bimodal (having two distinct peaks).
- 2 It indicates Central Tendency and Variation of data.
- 3 Detects Outliers

Histogram: Example -1

Setps for Histogram

- ① Construct frequency distribution with class boundaries
- ② Measure class boundaries(C.B) on horizontal axis with uniform interval.
- ③ Measure frequency (relative frequency, percentage frequency) on vertical axis.
- ④ Plot the data and draw bars.

Histogram: Example -1

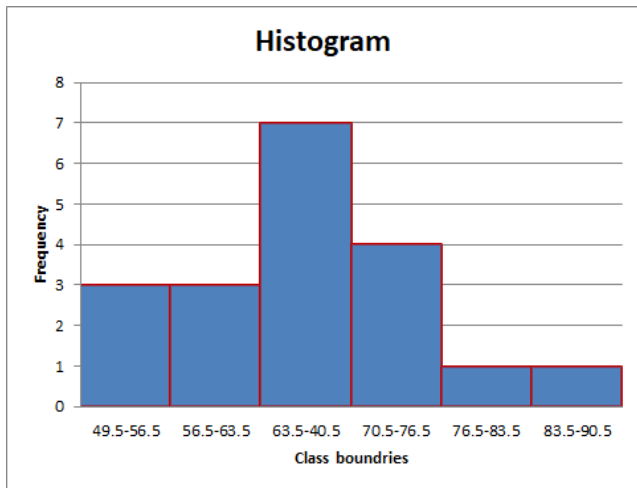
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Table: Frequency Distribution with C.B for Total Marks

Class Limits	Class Boundries	Frequency
50-56	49.5-56.5	3
57-63	56.5-63.5	3
64-70	63.5-40.5	7
71-76	70.5-76.5	4
77-83	76.5-83.5	1
84-90	83.5-90.5	1

Histogram: Example -1



Frequency Polygon: Example -1

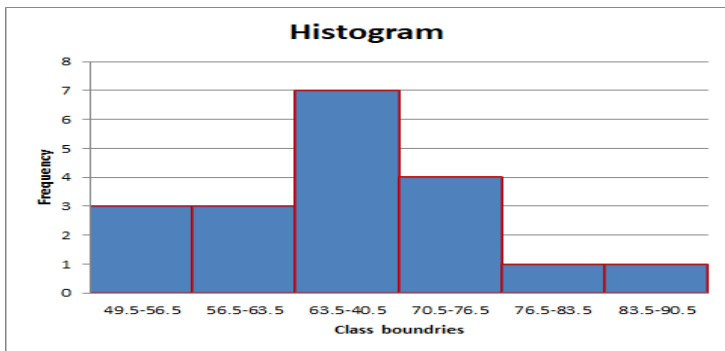
Steps:

(1)- Draw histogram

(2)- Identify mid point of each bar and join these points with straight lines

Draw frequency polygon using given Histogram drawn for Total Marks

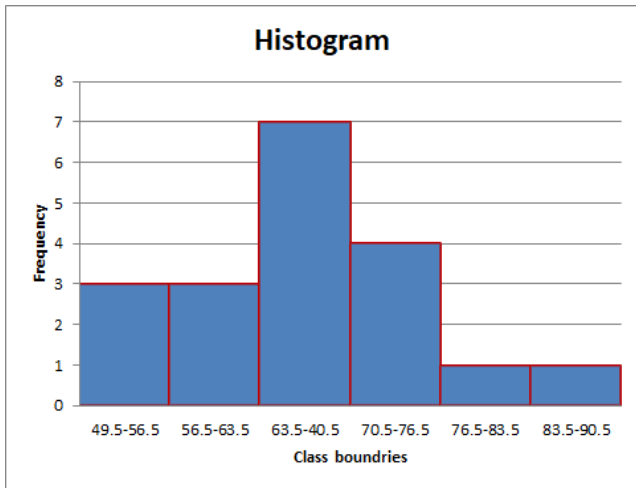
Figure:



Frequency Curve: Example -1

Draw frequency curve using given Histogram drawn for Total Marks

Figure:



Tabular methods

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Presentation of Qualitative Data

Tabular methods

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- 2 Percentage Frequency Distribution.

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- 3 Cross Tabulation (Contingency Table).

Graphical methods

- 1 Pie Chart.

Presentation of Qualitative Data

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Graphical methods

- ① Pie Chart.
- ② Simple Bar Chart

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Graphical methods

- ① Pie Chart.
- ② Simple Bar Chart
- ③ Multiple Bar Chart

Presentation of Qualitative Data

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Graphical methods

- 1 Pie Chart.
- 2 Simple Bar Chart
- 3 Multiple Bar Chart
- 4 Component Bar Chart etc

Example

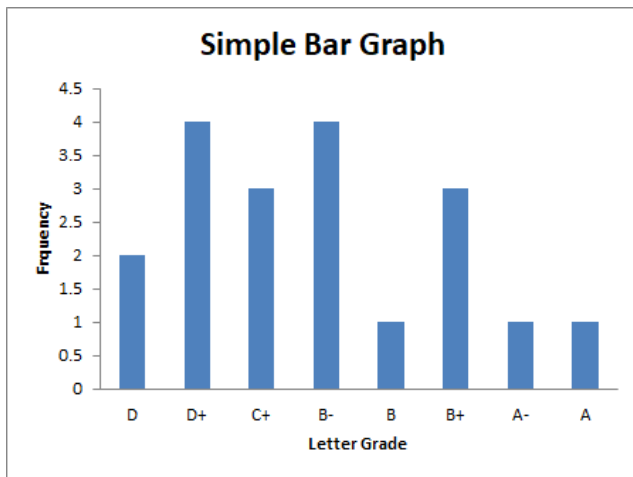
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Qualitative Frequency Distribution

Grade	Total
D	2
D+	4
C+	3
B-	4
B	1
B+	3
A-	1
A	1

Simple Bar Chart

A simple bar chart is used to represent and compare the values of one variable across different categories. Each category is represented by one bar.



Pie Chart

A pie chart is a circular graph divided into sectors (slices), where each sector represents a category's share of the whole.

- 1 To show how a total is divided among different categories.
- 2 To make it easy to compare the relative sizes or percentages of categories.
- 3 To present percentage distributions in an intuitive form.
- 4 To understand what components make up a total.
- 5 To present categorical data in a form that can be understood quickly.

Pie Chart

Grade	Total	defree	%F
D	2	38	10.526
D+	4	76	21.053
C+	3	57	15.789
B-	4	76	21.053
B	1	19	5.263
B+	3	57	15.789
A-	1	19	5.263
A	1	19	5.263

Pie Chart

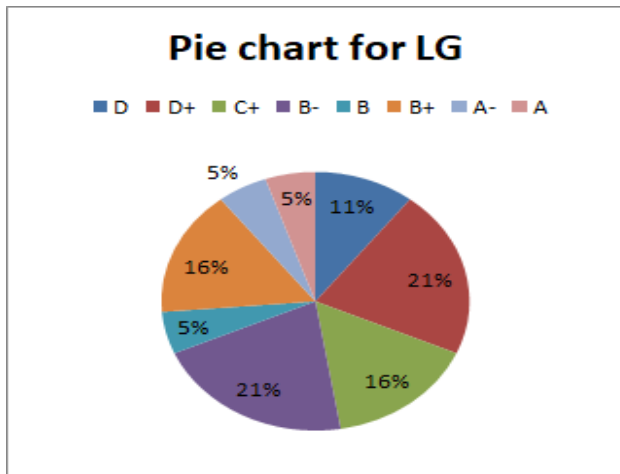


Figure:

Multiple Bar Chart

Table: Cross Tabulation

Grade	M	F
D	1	1
D+	4	0
C+	2	1
B-	2	2
B	1	0
B+	3	0
A-	1	0
A	0	1

Multiple Bar Chart

Use: To compare two or more related variables across the same categories.

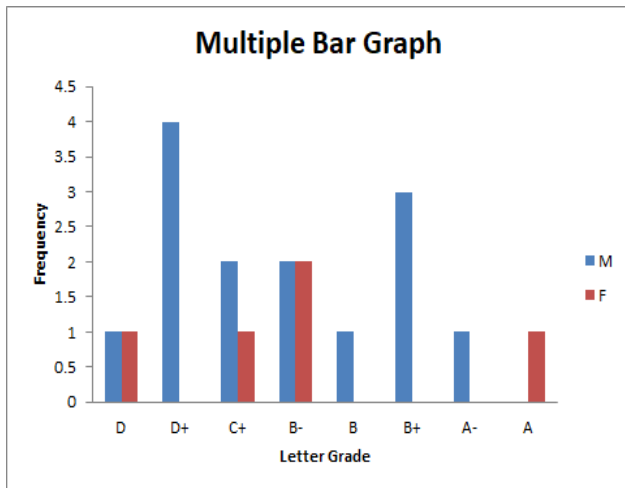


Figure:

Component Bar Chart

Grade	M	F	Total
D	1	1	2
D+	4	0	4
C+	2	1	3
B-	2	2	4
B	1	0	1
B+	3	0	3
A-	1	0	1
A	0	1	1

Component Bar Chart

Use: To show the total value and its components simultaneously.

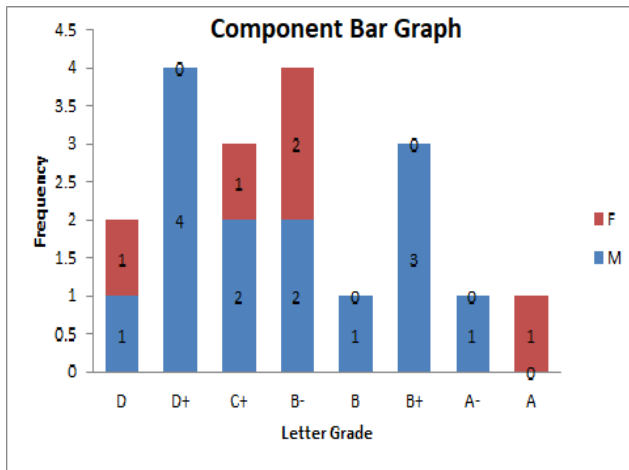
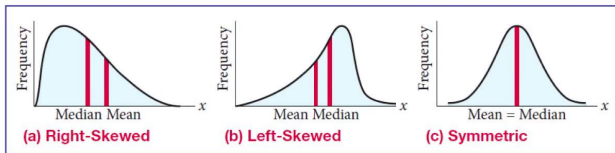


Figure:

Skewed and Symmetric Distributions

- Symmetric Data
 - Data sets whose values are evenly spread around the center.
- Skewed Data
 - Data sets that are not symmetric



Demerits of Presentation of Data

Although presentation of data in different forms (textual, Tabular or Graphical) make raw data more meaningful. But these methods fail to identify four major properties that describe any set of quantitative data. These four major properties are:

- ❶ **Central Tendency:** The numerical values of observation (also called central value) around which most numerical values of data show a tendency to cluster or group.

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- ③ **Skewness:** The extent of departure of values from symmetrical (Bell Shape) Distribution.

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- ④ **Kurtosis.** The extent of peakness of symmetrical distribution.

Descriptive Measure or Summary Measure

The four properties are used to extract and summarize major feature of the data set by the application of certain statistical methods called descriptive measures or summary measures: These Measures are:

- ➊ **Measures of Central Tendency:**
- ➋ **Measures of Variation or Dispersion:**
- ➌ **Measures of Skewness:**
- ➍ **Measures of Kurtosis.**

The End